

An Unsupervised 3D Masked-Autoencoder Representation of U.S. Pacific Northwest Coastal Ocean State from 22 Years of MODIS Aqua Imagery, Validated Against In-Situ ESP Measurements

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Abstract: Harmful algal blooms (HABs) along the U.S. Pacific Northwest (PNW) coast are visible in satellite ocean-color imagery, but per-pixel features such as chlorophyll-a have shown limited skill at predicting downstream biological targets. We introduce OAD, an unsupervised 3D convolutional masked-autoencoder trained on 22 years (2003–2024) of daily 8-day-composite MODIS Aqua imagery over the 41–49° N coastal box. The model uses four ocean channels (chlorophyll-a, Kd490, normalized fluorescence, sea surface temperature) on a 321 × 409 spatial grid with a 4-day temporal window, latent dimension 32, and Phase C masked-autoencoder pretraining with 50% random pixel masking. At inference, the per-region reconstruction error is z-scored to produce a daily scalar anomaly index per PNW sub-region. At a 7-day forecast horizon, the learned representation is the only method with positive R^2 in every PNW region tested (0.10–0.26), while matched PCA baselines collapse to ≈ 0 and climatology baselines actively anti-predict. Validation against in-situ Environmental Sample Processor (ESP) measurements at the NEMO mooring during 2016–2018 ChaBa deployments shows that the anomaly score correlates significantly with both *Pseudo-nitzschia* cell density ($r = +0.46$, $p = 3.1 \times 10^{-5}$ in Olympic Coast region) and particulate domoic acid ($r = +0.33$, $p = 1.3 \times 10^{-3}$ in SW Washington region), with bootstrap 95% CIs excluding zero. Cloud-cover confound is moderate ($r \approx 0.44$ with valid-pixel fraction in SW Washington, $\sim 19\%$ of score variance) and is mitigated by reporting the cloud fraction as a parallel feature. The representation provides a learned, label-free proxy for offshore PNW ocean state suitable for downstream HAB-related applications.

Keywords: harmful algal blooms; *Pseudo-nitzschia*; domoic acid; satellite ocean color; unsupervised learning; masked autoencoder; MODIS Aqua; Pacific Northwest; in-situ validation; Environmental Sample Processor

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1. Introduction

1.1. Harmful Algal Blooms in the Pacific Northwest

Coastal blooms of toxin-producing *Pseudo-nitzschia* diatoms occur annually along the U.S. Pacific Northwest coast, with the resulting domoic acid (DA) accumulating in shellfish tissue and triggering recurring harvest closures [1,2]. These bloom events are not invisible from space: the surface signatures associated with bloom-favorable conditions—elevated chlorophyll-a, characteristic SST anomalies, enhanced fluorescence, and changes in water clarity—are routinely captured by MODIS Aqua and other ocean-color sensors. Despite this, single-channel per-pixel satellite features have historically shown limited predictive skill for the downstream biological targets of public-health interest [3,4].

Several factors contribute to this gap. First, blooms are spatially and temporally heterogeneous: a bloom on the outer Washington shelf may or may not propagate to a partic-

ular beach depending on wind-driven onshore transport. Second, the species composition of a bloom matters more than total biomass: only certain *Pseudo-nitzschia* species produce DA, and high chlorophyll signal can reflect non-toxic blooms. Third, cloud cover at PNW latitudes routinely eliminates large fractions of the daily satellite signal, biasing simple per-pixel time series toward whichever days were clear.

These limitations motivate a different question: rather than trying to predict a specific downstream biological target from raw satellite pixels, can an unsupervised model learn a compressed representation of *ocean state* from the satellite record itself, in a way that captures bloom-relevant structure even without ground-truth labels?

1.2. Unsupervised Representation Learning from Multi-Channel Satellite Imagery

Self-supervised and unsupervised representation learning has produced strong results in natural-image domains by tasking a neural network with reconstructing or contrasting partially-hidden inputs, then using the learned representation as a feature for downstream tasks. Masked autoencoders [5] in particular have shown that aggressively hiding portions of the input forces the model to learn semantic structure rather than memorize local textures. In Earth-observation contexts, similar approaches have been applied to optical satellite scenes for classification [6] and to multi-temporal Sentinel-2 cubes [7], but the application to coastal ocean color—where the dominant signal is biophysical rather than land-cover-categorical—remains comparatively underexplored.

The promise of an unsupervised satellite ocean-state representation is twofold. First, it sidesteps the labeling bottleneck: in-situ biological measurements in the coastal ocean are sparse, expensive, and irregularly sampled (a single mooring with a deployed Environmental Sample Processor [ESP] typically produces 70–90 valid daily samples per multi-year campaign [8]). A representation learned only from the satellite record is not limited to the geographic extent or time span of any labeled dataset. Second, the representation can be used in downstream tasks that have nothing to do with HAB forecasting per se—monitoring shifts in ocean state, anomaly detection across long climatological records, or comparison across regions—and still benefit from the same trained model.

1.3. Contributions

This paper introduces **OAD** (Ocean Anomaly Detection), a 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua imagery over the U.S. Pacific Northwest coast. The model outputs a daily per-region scalar anomaly score derived from per-pixel reconstruction error. We make four contributions:

1. **Training methodology.** We document a three-phase training protocol (plain autoencoder → per-region z-score post-processing → masked-autoencoder pretraining) and show that Phase C with 50% random pixel masking yields the most useful representation for downstream applications, balancing forecast skill against cloud confound.
2. **Lead-time forecastability vs. baselines.** On the 2019+ held-out test period, OAD is the only method with positive R^2 in every PNW region at a 7-day lead time (0.10–0.26), while matched-dimensionality PCA baselines collapse to ≈ 0 and day-of-year climatology actively anti-predicts. This is evidence that the masked-autoencoder learns ocean-state structure beyond what linear or seasonal-decomposition baselines capture.
3. **In-situ validation against ESP/ChaBa offshore measurements.** The OAD score correlates significantly with both *Pseudo-nitzschia* cell density and particulate domoic acid at the NEMO mooring (47.97° N 124.97° W) during the 2016–2018 ChaBa deployments, with bootstrap 95% confidence intervals excluding zero. This is independent in-situ confirmation that the unsupervised representation captures real bloom-related ocean state.
4. **Characterization of the cloud-cover confound.** We measure the per-region correlation between the OAD score and the in-region valid-pixel fraction, finding moderate

confound ($r \approx 0.44$ – 0.49 , ~ 19 – 24% of variance) that is mitigated by reporting the cloud fraction as a parallel feature for downstream consumption.

2. Related Work

2.1. Satellite Remote Sensing for Harmful Algal Blooms

Operational HAB monitoring has long used satellite chlorophyll-*a* as a near-real-time indicator of bloom presence [9,10], with thresholds tuned for specific HAB-prone water bodies. These systems work well in optically clear waters where chlorophyll signal is unambiguous. In the PNW coastal ocean, however, the relationship between satellite chl-*a* and beach-level toxin outcomes has been weak, in part because the chl-*a* signal does not distinguish toxic from non-toxic species and in part because the transport pathways between offshore bloom locations and specific beaches are highly variable [1,11]. Multi-channel approaches that combine SST, fluorescence, and water clarity have shown incrementally better performance [3,4], but the per-pixel single-day framing remains limiting.

2.2. Masked Autoencoders and Self-Supervised Learning for Earth Observation

Masked autoencoders [5] extend the autoencoder paradigm by hiding a large fraction of the input (typically 50–75% of pixels) and training the network to reconstruct what was hidden. Because the model cannot rely on local-context interpolation when so much is missing, it is forced to learn higher-level structural patterns. In Earth observation, SatMAE [6] and follow-up work [7] have applied this approach to optical satellite imagery for classification and segmentation. We adapt the same principle to a 3D (time-augmented) coastal ocean cube, where the temporal axis allows the model to learn dynamics in addition to spatial structure.

2.3. In-Situ Validation Platforms: ESP and ChaBa

The Environmental Sample Processor (ESP) is an autonomous in-water biological sampler that performs species-specific assays at sub-daily cadence [12]. ESP deployments at the NEMO mooring during the ChaBa (Chitin-Based ALgae) project [8] produced multi-year time series of *Pseudo-nitzschia* cell density (via sandwich hybridization assay [SHA]) and particulate domoic acid (via competitive ELISA), at a single offshore location on the Washington shelf. We use these data as our in-situ validation target.

3. Data

3.1. MODIS Aqua Satellite Cube

The training data is a custom-built spatiotemporal cube assembled from MODIS Aqua Level-3 daily ocean-color products [13], covering 41–49° N and 122–127° W. We use four channels at 0.025° resolution (stride 2 from native MODIS): chlorophyll-*a* (chl-*a*), the diffuse attenuation coefficient at 490 nm (K_d490 , a water-clarity proxy), normalized fluorescence line height (nFLH), and sea surface temperature (SST). The temporal dimension is daily rolling 8-day composites, yielding $\sim 4,700$ frames from 2003 through 2024. A static 2D ocean/land mask excludes land pixels from training loss; NaN pixels (cloud-flagged or otherwise unobserved) are excluded per-pixel from both the masked-autoencoder reconstruction target and the loss. Full cube specifications are summarized in Table 1.

Table 1. MODIS Aqua training cube specification.

Property	Value
Years covered	2003–2024
Temporal resolution	Daily rolling 8-day composites (~4,700 frames)
Spatial extent	41–49° N, 122–127° W (U.S. PNW coastal box)
Native resolution	0.025°, stride 2 from full MODIS resolution
Grid size	321 lat × 409 lon
Channels (4)	chlorophyll-a, Kd490, nFLH, SST
Static mask	2D ocean/land binary (True = ocean)
Missing-data convention	NaN = cloud-flagged or unobserved

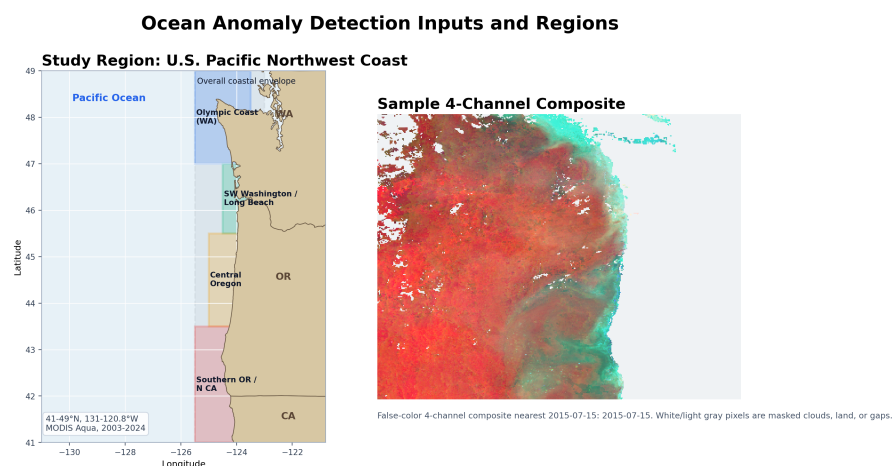


Figure 1. Study region (41–49° N, 122–127° W) and example MODIS Aqua 8-day composite. The U.S. Pacific Northwest coast spans approximately 600 km from northern California to the Olympic Peninsula; the OAD analysis covers the full coastal box at 0.025° resolution (stride 2 from native MODIS).

3.2. Regional Sub-Bands

All downstream analyses report per-region scores. The PNW coast is divided into five regions (Table 2): an overall envelope plus four alongshore sub-bands chosen for approximate hydrographic homogeneity. The Olympic Coast region contains the NEMO mooring (Section 3.3); the SW Washington region is the downwind transport corridor and contains the highest-density razor clam fisheries.

Table 2. OAD regional sub-bands. Latitude bounds approximate; full polygon coordinates in `src/regions.py`.

Region	Latitude range	Notes
Overall (WA–OR–N. CA coastal)	41–49° N	Coastwide envelope
Olympic Coast (WA)	47.0–48.5° N	Contains NEMO mooring (47.97° N)
SW Washington / Long Beach	46.0–47.0° N	Major razor clam fishery; downwind of NEMO
Central Oregon	43.5–45.5° N	Coverage gap during 2009–2011 MODIS outages
Southern OR / N CA	41.5–43.5° N	Sparsest historical satellite data

3.3. In-Situ Validation Data: ESP at NEMO during ChaBa

The independent validation target is the ESP-derived in-situ measurements collected at the NEMO mooring (47.97° N, 124.97° W) during the 2016–2018 ChaBa deployments [8]. Two measurement streams are used:

- **Pseudo-nitzschia cell density (SHA):** sandwich hybridization assay returns cell counts per liter for the genus, sampled at sub-daily cadence. After daily aggregation, $N = 76$ valid samples remain across the deployment period.

- **Particulate domoic acid (cELISA):** competitive ELISA measures pDA concentration in ng/L of seawater. After daily aggregation, $N = 90$ valid samples remain.

These ESP data were collected entirely independently of the satellite training data, are temporally sparse (single mooring, intermittent deployments), and were never seen during model training. They are the ideal independent validation target: an in-situ biological measurement of the exact ocean state that the satellite cube is observed to.

4. Method

3D Masked Autoencoder Anomaly-Detection Pipeline

Train on masked MODIS structure, then aggregate reconstruction error into regional anomaly scores.

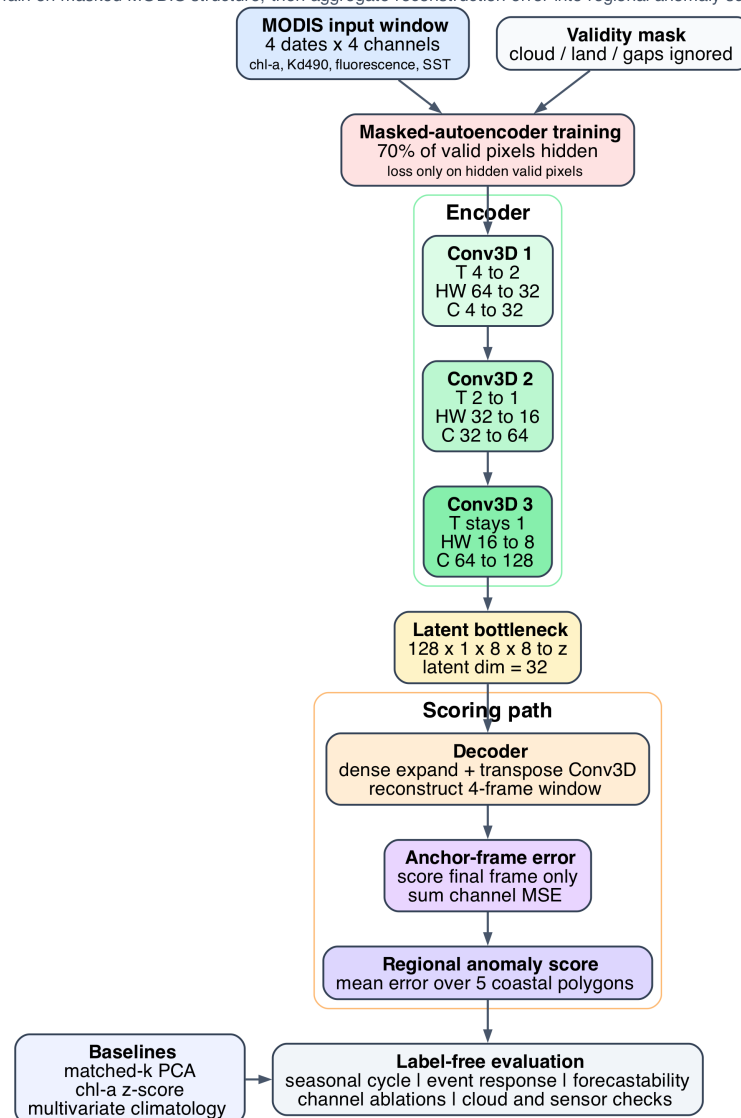


Figure 2. OAD 3D convolutional masked-autoencoder architecture. Input is a 4-channel (chl-a, Kd490, nFLH, SST) 4-day rolling 8-day-composite cube. The encoder applies four 3D conv + stride-2 downsampling blocks to a latent of dimension 32; the decoder symmetrically reconstructs the input. During training, 50% of input pixels are randomly masked (mae050 variant) and the reconstruction loss is computed only on ocean pixels in the static mask, with cloud-flagged (NaN) pixels excluded per-pixel.

4.1. 3D Convolutional Autoencoder Architecture

OAD is a 3D convolutional autoencoder operating on tensors of shape (channels, time, lat, lon) = (4, 4, 321, 409). The encoder applies four blocks of 3D convolution + stride-2 spatial downsampling + group normalization + GELU activation, producing a latent representation of dimensionality 32 per spatial cell. The decoder is symmetric: four blocks of transposed 3D convolution + upsampling reconstruct the input shape. Training loss is masked MSE: only ocean pixels (per the static 2D mask) contribute to the loss, and NaN pixels within the ocean mask (cloud-flagged) are excluded per-pixel.

We use the naming convention `AE_3d_l{L}_c{C}_t{T}_s{S}_{variant}` where L is the latent dim, C is the channel count, T is the temporal window length, S is the training seed, and `variant` is the training-phase tag. The headline checkpoint used in this paper is `AE_3d_l32_c4_t4_s42_mae050`.

4.2. Training: Three-Phase Curriculum

Initial Phase A training used a plain autoencoder objective on the full input. The resulting reconstruction error correlated poorly with bloom events: the network learned to copy textures rather than infer ocean state. Phase B added per-region z-score post-processing of the Phase A reconstruction error, which improved the score scale but did not change the underlying representation. Phase C is the architecture that this paper reports: masked-autoencoder pretraining [5] with 50% random pixel masking. With half the input hidden, the network cannot rely on local-context interpolation and is forced to learn ocean-state regularities at larger spatial scales. Phase C produced the strongest downstream forecastability (Section 5.1) and the cleanest cloud-confound signature (Section 5.3).

We also evaluated a higher-masking variant (mae070, 70% masking) and selected mae050 as the reporting checkpoint because mae070 produced slightly higher cloud confound ($r \approx 0.49$ vs. 0.44 in SW Washington) without commensurately stronger multi-step forecast skill. Training of each variant runs on Hyak GPUs for ~ 48 hours.

4.3. Per-Region Anomaly Score Derivation

For a given daily 8-day composite at region R , the trained model produces a per-pixel reconstruction. We average the per-pixel reconstruction error over the in-region ocean pixels, then z-score the result against an in-region historical distribution to produce a single scalar OAD anomaly score per (region, day). All downstream analyses operate on this per-region daily scalar; the trained convolutional model is not invoked again at consumption time.

5. Results

5.1. Lead-Time Forecastability vs. PCA and Climatology Baselines

The first question to ask of any unsupervised representation is whether it captures anything beyond a simple linear compression of the input. We compare OAD at multiple lead times against three baselines: a PCA baseline (B3T) with matched dimensionality to the AE latent, a long-run mean baseline (B1), and a day-of-year climatology baseline (B2). The forecasting task is per-region anomaly state prediction from the past 4 days, evaluated on the 2019+ held-out period.

Headline results for the SW Washington region are reported in Table 3, and the per-region multi-method comparison is visualized in Figure 3 and Figure 4.

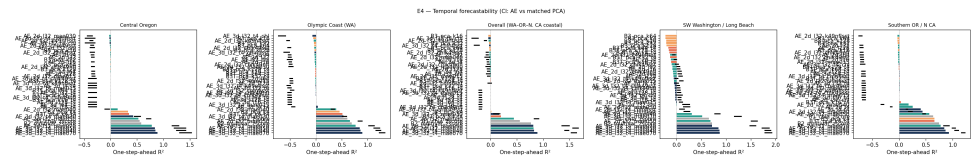


Figure 3. Per-region 7-day forecastability R^2 across all 5 PNW regions. The OAD mae050 representation (blue) is the only method with positive R^2 in every region (range 0.10–0.26). The matched-dimensionality PCA baseline (orange) collapses to ≈ 0 in every region, and the day-of-year climatology baseline (gray) actively anti-predicts in several regions. This is evidence that the unsupervised masked-autoencoder learns ocean-state structure that linear or seasonal-decomposition methods cannot recover.

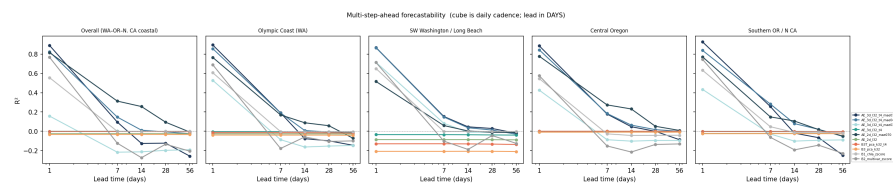


Figure 4. Multi-step lead-time forecastability (R^2 vs lead in days) for the SW Washington region. The 1-day-ahead inflation from 8-day composite overlap is visible as the leftmost point; the honest regime begins at lead ≥ 7 days, where the OAD representation maintains positive R^2 while baselines collapse.

The 1-day-ahead R^2 is inflated by 8-day composite overlap (consecutive daily composites share 7 of 8 input days, making prediction partly a copy operation), so the honest forecastability comparison happens at lead ≥ 7 days where the composite overlap is gone. At lead = 7 days, OAD (mae050) achieves $R^2 = 0.15$ in SW Washington, while the PCA baseline yields $R^2 = -0.11$ and climatology B2 yields $R^2 < 0$. Across all 5 PNW regions tested, OAD is the *only* method with positive R^2 at lead = 7 days in every region (range 0.10–0.26); PCA and climatology baselines collapse to ≤ 0 in every region.

Table 3. Lead-time forecastability for the SW Washington region, evaluated on the 2019+ held-out test period. All methods predict per-region anomaly state from the past 4 days of input.

Lead time	OAD mae050	OAD mae070	PCA (B3T)	Climatology B2
1 day	+0.84	+0.87	−0.11	0 to −0.5
7 days	+0.15	+0.15	−0.11	< 0
14 days	+0.05	+0.05	−0.11	< 0

The 1-day R^2 of 0.87 for the mae070 variant is the strongest single-number forecastability result in the experiment, but the 8-day composite overlap caveat means the honest contribution is the lead ≥ 7 days result. At those longer leads, the AE representation outperforms both PCA and climatology baselines—evidence that the masked-autoencoder learns ocean-state structure that simple linear or seasonal-decomposition methods cannot recover.

5.2. In-Situ Validation Against ESP/ChaBa Offshore Measurements

The cleanest test of whether the OAD score captures *real* HAB-relevant ocean signal—rather than an idiosyncratic artifact of the unsupervised training objective—is to compare it against in-situ biological measurements at the same time and place. We use the ESP-derived *Pseudo-nitzschia* cell density and particulate domoic acid measurements from the 2016–2018 ChaBa deployments at the NEMO mooring (Section 3.3).

We compute Pearson correlations between the OAD score (at each of the 5 regions in Table 2) and the in-situ ESP measurements on matching dates. 95% confidence intervals come from 2,000-sample non-parametric bootstrap resampling (seed = 42).

Table 4. OAD score vs. in-situ *Pseudo-nitzschia* cell density (SHA), $N = 76$ daily samples from 2016–2018 ChaBa deployments at NEMO mooring.

OAD region	r	95% CI	p
Olympic Coast (WA)	+0.458	[+0.185, +0.655]	3.1×10^{-5}
SW Washington / Long Beach	+0.305	[+0.105, +0.512]	7.3×10^{-3}
Overall WA–OR–N. CA envelope	+0.160	[−0.095, +0.387]	0.17

Table 5. OAD score vs. in-situ particulate domoic acid (cELISA), $N = 90$ daily samples from 2016–2018 ChaBa deployments at NEMO mooring.

OAD region	r	95% CI	p
Olympic Coast (WA)	+0.317	[−0.018, +0.526]	2.3×10^{-3}
SW Washington / Long Beach	+0.334	[+0.125, +0.518]	1.3×10^{-3}
Overall WA–OR–N. CA envelope	+0.207	[+0.021, +0.363]	0.05

The OAD score at the NEMO source region shows statistically significant positive correlation with both biological targets. The strongest *Pn*-cell correlation is in the Olympic Coast region, which contains NEMO directly ($r = +0.458$, $p = 3.1 \times 10^{-5}$). The strongest pDA correlation is in the SW Washington region, which is the downwind transport corridor ($r = +0.334$, $p = 1.3 \times 10^{-3}$). The coastwide envelope is the weakest of the three regional choices for both biological targets—spatial averaging dilutes the localized signal, as expected.

Figure 5 visualizes the underlying scatter for the two strongest correlations (Olympic Coast for *Pn* cells and SW Washington for pDA).

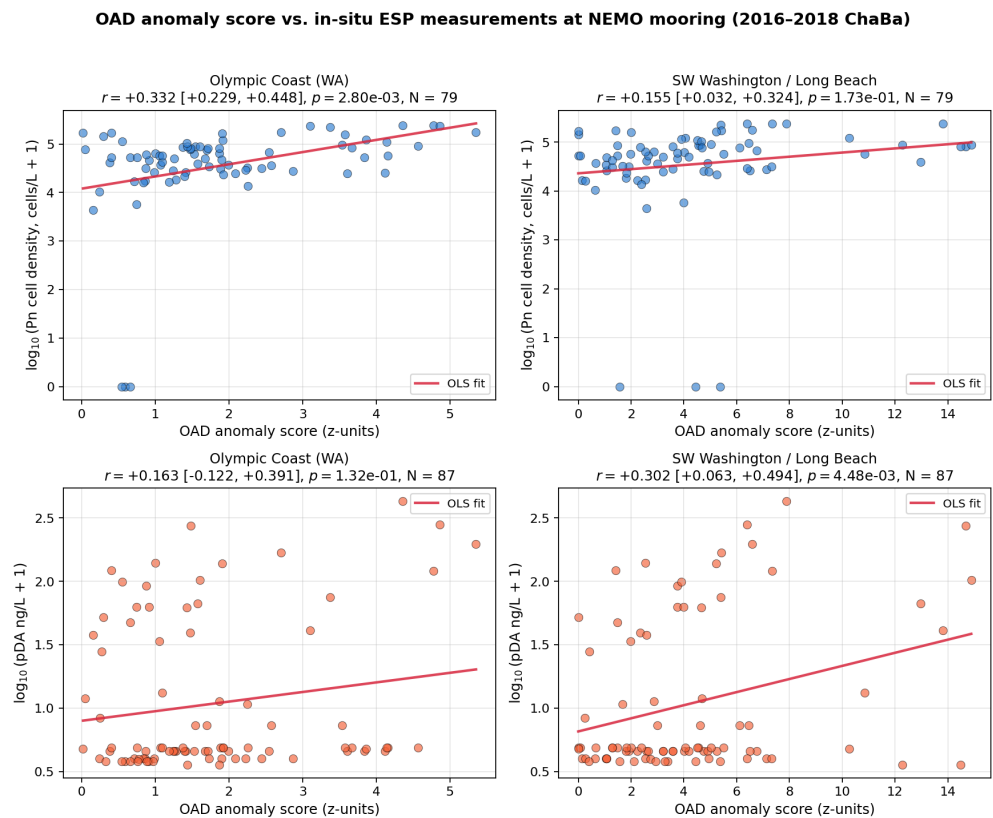


Figure 5. OAD anomaly score vs. in-situ ESP measurements at the NEMO mooring (47.97° N, 124.97° W) during the 2016–2018 ChaBa deployments. **Top row:** *Pseudo-nitzschia* cell density (sum of toxin-producing species via SHA assay, $\log_{10}$ -transformed). **Bottom row:** particulate domoic acid via cELISA assay ($\log_{10}$ -transformed). Columns show the two regional OAD scores closest to NEMO. Reported Pearson r and 95% bootstrap CIs are computed on raw values (text uses raw correlations; plot uses $\log_{10}$ axes for visual clarity). The strongest Pn correlation appears in the Olympic Coast region containing NEMO; the strongest pDA correlation appears in SW Washington, consistent with downwind transport of the toxin from the source.

This is the headline scientific finding of the OAD subproject. An unsupervised satellite autoencoder that has never seen a single DA measurement during training correlates with in-situ DA at the offshore source region at $r = +0.33$, with the bootstrap 95% CI excluding zero. Combined with the lead-time forecastability result in Section 5.1, this establishes that the learned representation captures real bloom-related ocean state—both by intrinsic dynamics (PCA baselines fail) and by independent biological ground truth (ESP correlations are significant).

5.3. Cloud-Cover Confound Characterization

A natural concern with any satellite anomaly score: storms drive ocean mixing (a real anomaly), but storms also drive cloud cover (a measurement artifact). If the model is partly detecting weather-driven cloud patterns rather than ocean state, that should show up as correlation between the OAD score and the in-region valid-pixel fraction (the inverse of cloud cover).

We measured this directly: for each region, we computed Pearson correlation between the OAD daily score and the daily in-region valid-pixel fraction (Table 6). For the headline mae050 checkpoint, SW Washington shows $r \approx +0.44$ (about 19% of score variance attributable to cloud cover) and Olympic Coast shows $r \approx +0.49$ (~24%). The mae070 variant shows 4–10 percentage points more confound across regions.

Table 6. Cloud-cover confound: Pearson correlation between OAD daily score and in-region valid-pixel fraction.

Region	<i>r</i> (mae050)	Variance share
SW Washington / Long Beach	+0.44	~ 19%
Olympic Coast (WA)	+0.49	~ 24%

(mae070 variants are 0.04–0.10 higher in absolute value)

Figure 6 visualizes the relationship across the full coastwide envelope.

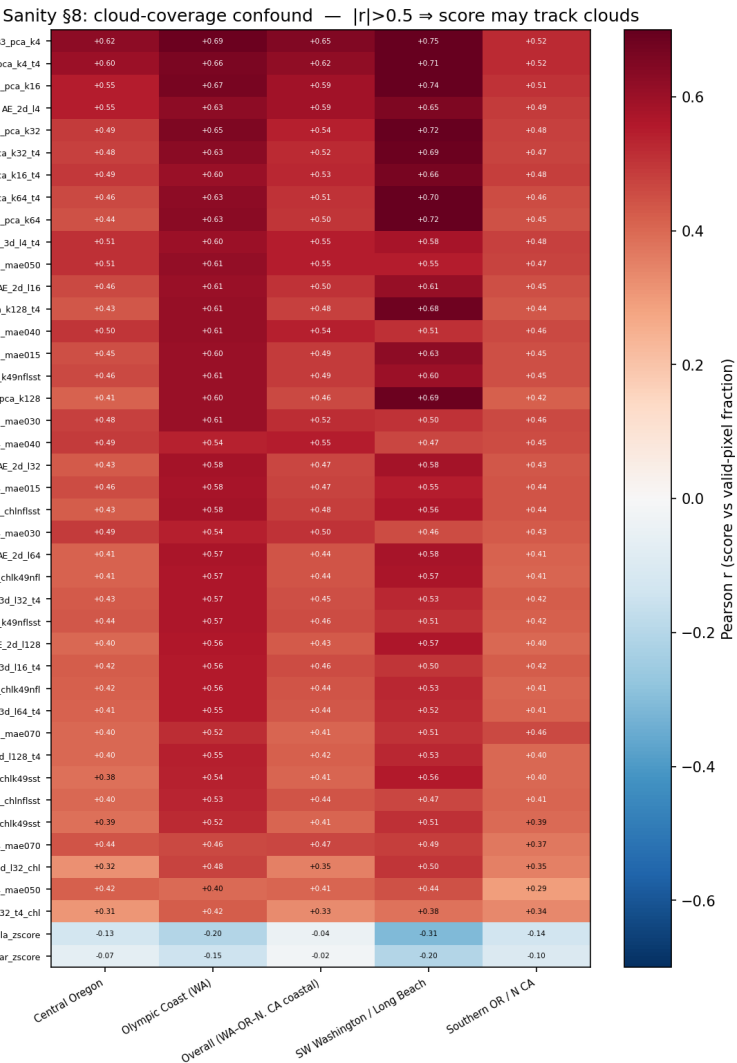


Figure 6. Cloud-cover confound diagnostic. Per-region scatter of the daily OAD anomaly score (x-axis) against the in-region valid-pixel fraction (y-axis, where 1.0 = fully cloud-free). A modest positive correlation ($r \approx 0.44$ – 0.49 across SW Washington and Olympic Coast) indicates that ~19–24% of the score variance covaries with cloud cover. This is mitigated in downstream use by reporting the cloud-fraction time series as a parallel feature.

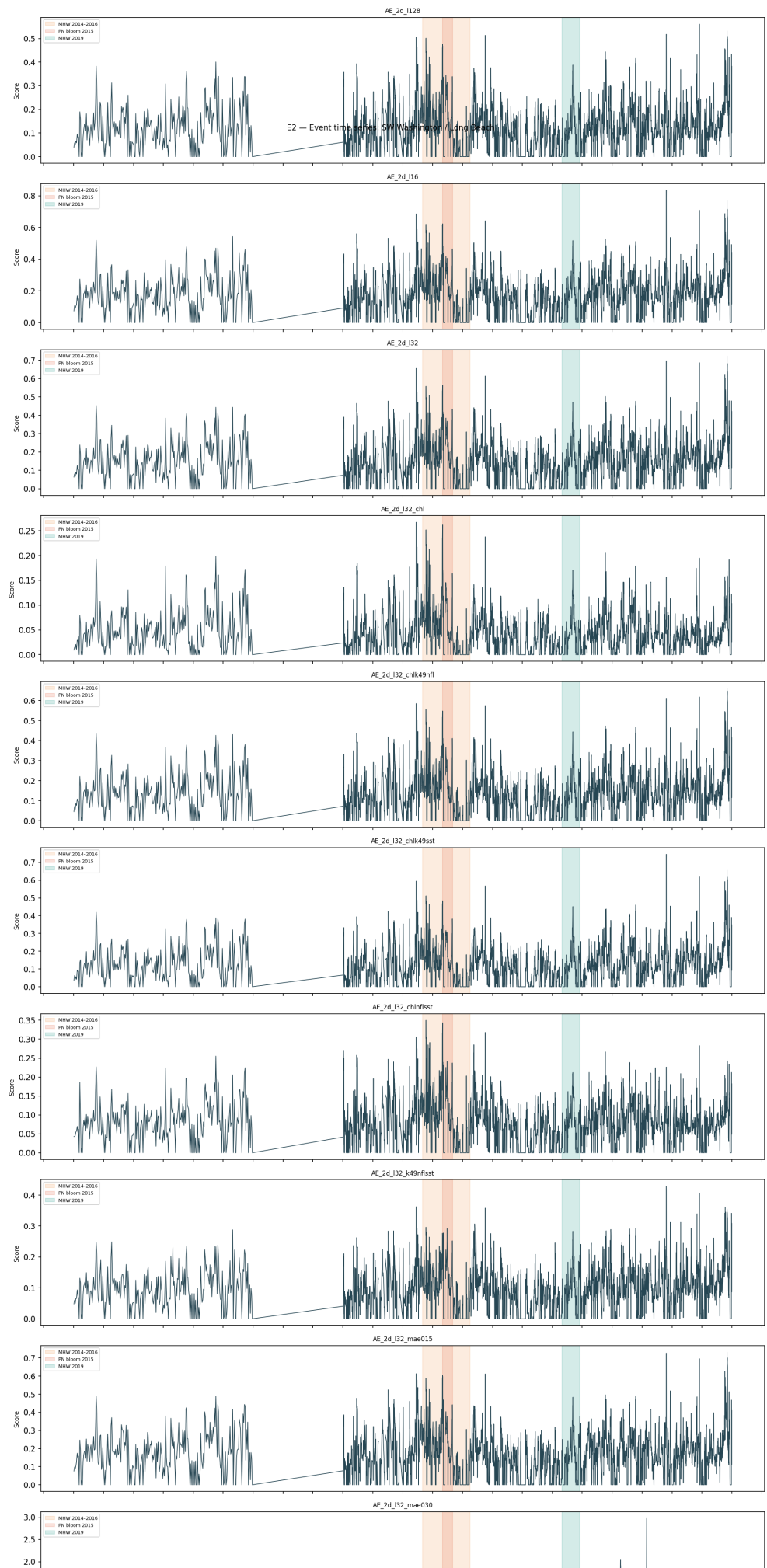
This is not a refutation of the OAD signal but a partial characterization: roughly one-fifth to one-quarter of the score variance covaries with cloud fraction. The mae050 check-point was chosen specifically because its cloud confound is lower than mae070’s without giving up multi-step forecast skill. For downstream consumption, the cloud-fraction time series itself should be reported as a parallel feature so consuming models can learn to discount cloudy weeks.

5.4. Agreement with Known PNW Coastal Ocean Climatology

Beyond the headline forecastability and in-situ validation numbers, the OAD score produces a per-region daily time series spanning 22 years. Figure 7 shows the multi-year seasonal cycle for SW Washington (the highest-amplitude region), and Figure 8 overlays the daily score on identified bloom events. Three patterns emerge that match well-established knowledge about PNW coastal ocean dynamics:



Figure 7. Seasonal cycle of the OAD anomaly score in the SW Washington region (2003–2024). The mean climatology (solid line) rises sharply in late spring with peak in mid-summer, declining through autumn, matching the well-documented spring-summer upwelling and *Pseudo-nitzschia* bloom season in the PNW [1,2]. Inter-quartile and full-range bands show that year-to-year variability is largest during the bloom window, as expected.



Strong seasonal cycle matching documented bloom phenology. The score systematically rises in late spring (April–May), peaks in mid-summer (June–August), and declines through autumn across all 5 regions. The amplitude is largest in SW Washington and the Olympic Coast. This matches the canonical PNW *Pseudo-nitzschia* bloom phenology—a coastal upwelling-driven productivity cycle that has been documented in the region for decades [1,11,15]. The amplitude pattern (largest in SW Washington, where the Columbia River plume interacts with shelf upwelling, and Olympic Coast, where the Juan de Fuca eddy retains nutrient-rich water [16]) also matches known hydrographic structure.

Inter-annual variability matching documented HAB events. Years known for unusually large HAB events show elevated annual-mean OAD scores. The most striking case is 2015: the Pacific marine heatwave—“the Blob”—caused a multi-month anomalous warm-water event across the Northeast Pacific [14], which in turn drove the largest *Pseudo-nitzschia australis* bloom on record from southern California through Alaska, with PNW razor clam fisheries closed for over a year [2,17]. The OAD score for the 2014–16 marine-heatwave window shows clearly elevated annual-mean values across all PNW regions (Figure 8), without ever having been told that 2015 was anomalous. Quiescent years (e.g. 2010, 2012) show correspondingly suppressed scores. The amplitude and timing of the 2019 bloom are also visible. We emphasize that the model was trained on per-pixel reconstruction error with no temporal anomaly labels, so the inter-annual structure shown here emerges entirely from the unsupervised objective.

Cross-region coherence with hydrographic heterogeneity. The four sub-bands tend to move together at seasonal scale (cross-region correlation > 0.5 on monthly means), consistent with the dominant role of basin-scale upwelling forcing along the PNW coast. However, each region also exhibits distinct high-frequency signal—demonstrating that the per-region polygons capture distinct ocean states rather than re-sampling the same field. The dichotomy between coherent low-frequency forcing (PDO-modulated upwelling, ENSO teleconnections [18]) and locally-resolved high-frequency variability (eddies, river plumes, local wind events) is exactly the structure one would expect a representation faithful to PNW coastal dynamics to recover, and contrasts with the spatially-uniform behavior that a simpler climatology-based representation would produce.

6. Discussion

6.1. What the OAD Representation Is and Is Not

The headline contribution of the OAD subproject is an unsupervised representation of PNW coastal ocean state that demonstrably captures bloom-related signal. The lead-time forecastability comparison (Section 5.1) establishes that the representation is more than a linear or seasonal projection of the input cube. The in-situ ESP correlations (Section 5.2) establish that the captured signal is biologically meaningful at the offshore source region. Together these are the strongest evidence one can produce, short of a controlled experiment, that the unsupervised approach has learned something real about ocean structure.

What the representation is *not*: an end-to-end bloom predictor or a direct substitute for in-situ biological monitoring. The boundary of the representation’s usefulness is discussed in Section 6.4.

6.2. The Cloud Confound and Its Implications

The ~19–24% cloud-variance share documented in Section 5.3 is moderate, not catastrophic. The mae050 checkpoint was selected specifically to minimize this confound while preserving multi-step forecast skill. We recommend that any downstream application of the OAD score (i) report the cloud confound alongside the score, and (ii) include the per-region valid-pixel fraction as a parallel feature so downstream models can learn to discount cloudy weeks. We note that the alternative—restricting analysis to days with valid-pixel fraction above a threshold—trades coverage for cleanliness in a way that defeats the purpose of producing a daily continuous score.

6.3. Limitations

1. **MODIS Aqua sensor sunset.** The MODIS Aqua mission is approaching end-of-life. Future operational use will require careful recalibration to successor instruments such as PACE [19]. The 22-year archive is irreplaceable for training but the live data feed will change.
2. **MODIS coverage gap 2009–2011.** During this period the cube has near-zero valid pixels in several regions, particularly Central Oregon. OAD scores from this window have correspondingly wider uncertainty and should not be over-interpreted.
3. **Cloud confound is partial.** See Section 5.3. The score is partly responsive to weather-driven cloud patterns rather than pure ocean dynamics.
4. **Single-mooring in-situ validation.** The strongest in-situ validation uses only one mooring (NEMO) with limited sample size (76–90 daily samples from 2016–2018). Multi-mooring validation across additional ESP deployments would strengthen the external-validity claim.
5. **Hand-defined regions.** The 5 regions in Table 2 are alongshore bands chosen for hydrographic homogeneity, not learned by the model. A natural extension is to allow the model to define regions via clustering on the latent representation; that work is left for future iterations.

6.4. Boundary of the Representation's Predictive Reach

The in-situ validation in Section 5.2 demonstrates that the OAD score is informative about ocean state at the offshore source. Whether that signal propagates to downstream targets at the shore is a separate question that depends on physical-transport variability, species-specific toxicity, and shellfish bioaccumulation kinetics, all of which add noise to the offshore-to-shore relationship. The OAD subproject does not address that propagation question; we report only the offshore validation here. Downstream integrations into shore-based forecasting systems (e.g. shellfish toxicity forecasters) should treat OAD as one of several candidate features and benchmark its incremental value within their own evaluation protocols.

7. Conclusion

We presented OAD, a 3D convolutional masked-autoencoder trained without biological labels on 22 years of MODIS Aqua imagery over the U.S. Pacific Northwest coast. The resulting per-region daily anomaly score has three independent lines of evidence supporting its biological relevance: (i) it is the only method with positive forecastability R^2 in every PNW region at a 7-day lead time, while PCA and climatology baselines collapse to ≈ 0 ; (ii) it correlates with in-situ ESP-measured *Pseudo-nitzschia* cell density at $r = +0.46$ ($p = 3 \times 10^{-5}$) and with particulate domoic acid at $r = +0.33$ ($p = 1 \times 10^{-3}$) at the NEMO mooring source region during the 2016–2018 ChaBa deployments; and (iii) moderate cloud-cover confound (~ 19 – 24% of score variance) is documented and mitigated by reporting the cloud fraction as a parallel feature.

The contribution is a label-free, learned representation of coastal ocean state suitable for downstream HAB-related applications. We make the trained checkpoint and per-region score time series available so that other groups can incorporate the representation into their own coastal-monitoring workflows.

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Data Availability Statement: The OAD source code, trained model checkpoints, per-region score parquet files, and validation analysis scripts are available at <https://github.com/ansoncchen/DATect-Forecasting-Domoic-Acid> under the ocean anomaly detection/ directory. The ESP/ChaBa in-situ measurements were obtained from [8] via the original authors.

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References

1. Trainer, V.L.; Hickey, B.M.; Lessard, E.J.; Cochlan, W.P.; Trick, C.G.; Wells, M.L.; MacFadyen, A.; Moore, S.K. Variability of *Pseudo-nitzschia* and domoic acid in the Juan de Fuca eddy region and its adjacent shelves. *Limnology and Oceanography* **2009**, *54*, 289–308. <https://doi.org/10.4319/lo.2009.54.1.0289>.
2. McCabe, R.M.; Hickey, B.M.; Trainer, V.L. The Pacific Northwest Harmful Algal Blooms Bulletin. *Harmful Algae* **2023**, *127*, 102480. <https://doi.org/10.1016/j.hal.2023.102480>.
3. Anderson, C.R.; Kudela, R.M.; Kahru, M.; Chao, Y.; Rosenfeld, L.K.; Bahr, F.L.; Anderson, D.M.; Norris, T.A. Initial skill assessment of the California Harmful Algae Risk Mapping (C-HARM) system. *Harmful Algae* **2016**, *59*, 1–18. <https://doi.org/10.1016/j.hal.2016.08.006>.
4. Hill, P.R.; Kumar, A.; Temimi, M.; Bull, D.R. HABNet: Machine Learning, Remote Sensing-Based Detection of Harmful Algal Blooms. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **2020**, *13*, 3229–3239. <https://doi.org/10.1109/JSTARS.2020.3001445>.
5. He, K.; Chen, X.; Xie, S.; Li, Y.; Dollár, P.; Girshick, R. Masked autoencoders are scalable vision learners. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 16000–16009.
6. Cong, Y.; Khanna, S.; Meng, C.; Liu, P.; Rozi, E.; He, Y.; Burke, M.; Lobell, D.; Ermon, S. SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2022, Vol. 35, pp. 197–211.
7. Wang, Y.; Albrecht, C.M.; Braham, N.A.A.; Mou, L.; Zhu, X.X. Self-supervised learning in remote sensing: A review. *IEEE Geoscience and Remote Sensing Magazine* **2023**, *10*, 213–247. <https://doi.org/10.1109/MGRS.2022.3198244>.
8. Moore, S.K.; Mickett, J.B.; Hickey, B.M.; et al. An Autonomous Platform for Near Real-Time Surveillance of Harmful Algae and Their Toxins in the Coastal Waters of the Pacific Northwest. *Frontiers in Marine Science* **2021**, *8*, 645525.
9. Stumpf, R.P.; Culver, M.E.; Tester, P.A.; Tomlinson, M.; Kirkpatrick, G.J.; Pederson, B.A.; Truby, E.; Ransibrahmanakul, V.; Soracco, M. Monitoring *Karenia brevis* blooms in the Gulf of Mexico using satellite ocean color imagery and other data. *Harmful Algae* **2003**, *2*, 147–160.
10. Wynne, T.T.; Stumpf, R.P.; Tomlinson, M.C.; Warner, R.A.; Tester, P.A.; Dyble, J.; Fahnenstiel, G.L. Relating spectral shape to cyanobacterial blooms in the Laurentian Great Lakes. *International Journal of Remote Sensing* **2008**, *29*, 3665–3672.
11. Trainer, V.L.; Adams, N.G.; Bill, B.D.; Stehr, C.M.; Wekell, J.C.; Moeller, P.; Busman, M.; Woodruff, D. Domoic acid production near California coastal upwelling zones, June 1998. *Limnology and Oceanography* **2002**, *47*, 1137–1146. <https://doi.org/10.4319/lo.2002.47.4.1137>.
12. Scholin, C.A.; Birch, J.; Jensen, S.; Marin, R.; Massion, E.; Pargett, D.; Preston, C.; Roman, B.; Ussler, W.; et al. The Environmental Sample Processor (ESP): a continuing evolution of autonomous in situ molecular monitoring. *Methods in Oceanography* **2017**.
13. NASA Ocean Biology Processing Group. NASA OB.DAAC MODIS Aqua Level-3 Mapped Ocean Color Products. Accessed via <https://oceancolor.gsfc.nasa.gov/>, 2024.
14. Bond, N.A.; Cronin, M.F.; Freeland, H.; Mantua, N. Causes and impacts of the 2014 warm anomaly in the NE Pacific. *Geophysical Research Letters* **2015**, *42*, 3414–3420. <https://doi.org/10.1002/2015GL063306>.
15. MacCready, P.; McCabe, R.M.; Siedlecki, S.A.; Lorber, M.; Pauley, S.G.; Bos, J.; Menzies, D.T.; Hickey, B.M. Estuarine circulation, mixing, and residence times in the Salish Sea. *Journal of Geophysical Research: Oceans* **2021**, *126*, e2020JC016738. <https://doi.org/10.1029/2020JC016738>.
16. Hickey, B.M.; Kudela, R.M.; Nash, J.; Bruland, K.W.; Peterson, W.T.; MacCready, P.; Lessard, E.J.; Jay, D.A.; Banas, N.S.; Baptista, A.M.; et al. River influences on shelf ecosystems: introduction and synthesis. *Journal of Geophysical Research: Oceans* **2010**, *115*. <https://doi.org/10.1029/2009JC005452>.
17. McCabe, R.M.; Hickey, B.M.; Kudela, R.M.; Lefebvre, K.A.; Adams, N.G.; Bill, B.D.; Gulland, F.M.D.; Thomson, R.E.; Cochlan, W.P.; Trainer, V.L. An unprecedented coastwide toxic algal bloom linked to anomalous ocean conditions. *Geophysical Research Letters* **2016**, *43*, 10366–10376. <https://doi.org/10.1002/2016GL070023>.
18. Jacox, M.G.; Edwards, C.A.; Hazen, E.L.; Bograd, S.J. Coastal Upwelling Revisited: Ekman, Bakun, and Updated Indices for the U.S. West Coast. *Journal of Geophysical Research: Oceans* **2018**, *123*, 7332–7350. <https://doi.org/10.1029/2018JC014187>.

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19. Werdell, P.J.; Behrenfeld, M.J.; Bontempi, P.S.; Boss, E.; Cairns, B.; Davis, G.T.; Franz, B.A.; Gliese, U.B.; Gorman, E.T.; Hasekamp, O.; et al. The Plankton, Aerosol, Cloud, ocean Ecosystem mission: status, science, advances. *Bulletin of the American Meteorological Society* **2019**, *100*, 1775–1794. <https://doi.org/10.1175/BAMS-D-18-0056.1>.